NUCLEAR POWER CORPORATION OF INDIA LIMITED

WELDING PROCEDURE SPECIFICATION (WPS) - ASME QW-482

SECTION 1 - IDENTIFICATION			
WPS NO.	NPCIL-STD-SM-88	REVISION NO.	R-1
DATE	2014-07-07	QUALIFICATION STATUS	Qualified
SUPPORTING PQR NO.(S)	D		

QA APPROVED

SECTION 2 - WELDING PROCESSES			
WELDING PROCESS (ROOT)	SMAW	WELDING PROCESS (FILL)	SMAW
TYPE	Manual	JOINT DESIGN	Groove
BACKING	With or Without	BACKING MATERIAL	As per design document

SECTION 4 - BASE METALS (QW-403)			
P-NO.	P-No. 8 Group 1	TO P-NO.	P-No. 8 Group 1
MATERIAL SPEC / GRADE	Synthetic but code-style material family		
THICKNESS RANGE (GROOVE)	1.5 mm to 200 mm	THICKNESS RANGE (FILLET)	All Thickness

SECTION 5 - POSITIONS (QW-405)			
POSITION(S) OF GROOVE	All	WELDING PROGRESSION	Vertical Uphill
POSITION(S) OF FILLET	All	PWHT	Not Required

PREPARED BY | REVIEWED BY (QA) | APPROVED BY

NUCLEAR POWER CORPORATION OF INDIA LIMITED

WPS EXECUTION PARAMETERS — NPCIL-STD-SM-88

SECTION 6 - PREHEAT & PWHT (QW-406 / QW-407)			
PREHEAT TEMP, MIN	10.0 °C	INTERPASS TEMP, MAX	150.0
PWHT REQUIREMENTS	Not Required		

SECTION 8 - ELECTRICAL CHARACTERISTICS (QW-409)	
CURRENT / POLARITY	DC-EP
TRAVEL SPEED RANGE	40-80 mm/min

SECTION 9 - TECHNIQUE (QW-410)	
STRING OR WEAVE BEAD	Root: Stringer / Fill: Weave
CLEANING METHOD	Brushing or Grinding
PEENING ALLOWED	False

SECTION 10 - FILLER METALS (QW-404)			
SFA SPECIFICATION	5.4	AWS CLASSIFICATION	E308L-16
F-NO.	varies	A-NO.	varies

NOTES & RESTRICTIONS:

Synthetic WPS record used for contrast and edge cases.

JOINT SCHEMATIC